AutomataStudio — Finite Automat
https://reyescarlata0.github.io/automata-studio/?utm_source=chatgpt.com
Chat
Extensions
Other favorites

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ?

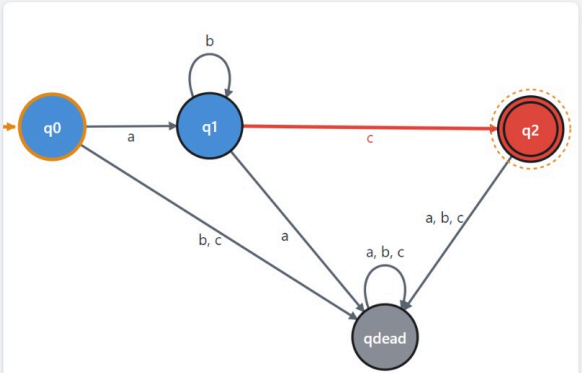
States | Alphabet | Transitions

FROM: qdead
SYMBOL: c
TO: qdead
Add transition

- q1 —b→ q1
- q1 —c→ q2
- q1 —a→ qdead
- q2 —a→ qdead
- q2 —b→ qdead
- q2 —c→ qdead
- qdead —a→ qdead
- qdead —b→ qdead

Delete transition

DFA: binary strings ending in 1 — DFA



State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

Deterministic (DFA)	Yes
Complete	Yes
Connected	Yes
ε-transitions	No
States / symbols	4 / 3

Unreachable states

None

Dead states

qdead

Missing transitions

None — the automaton is complete.

Validation

The automaton is well formed.

INPUT STRING: abbbbc | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

a b b b b c

Step 6 / 6 — On 'c': {q1} → {q2}

ACCEPTED

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

FROM
qdead

SYMBOL
c

TO
qdead

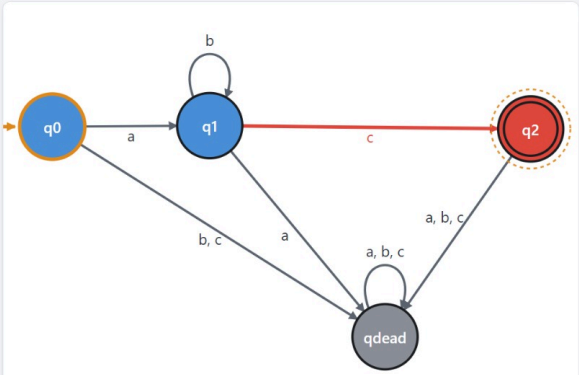
Add transition

q1 —b→ q1
q1 —c→ q2
q1 —a→ qdead
q2 —a→ qdead
q2 —b→ qdead
q2 —c→ qdead
qdead —a→ qdead
qdead —b→ qdead

Delete transition

DFA: binary strings ending in 1

DFA



State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

Deterministic (DFA)	Yes
Complete	Yes
Connected	Yes
ε-transitions	No
States / symbols	4 / 3

Unreachable states

None

Dead states

qdead

Missing transitions

None — the automaton is complete.

Validation

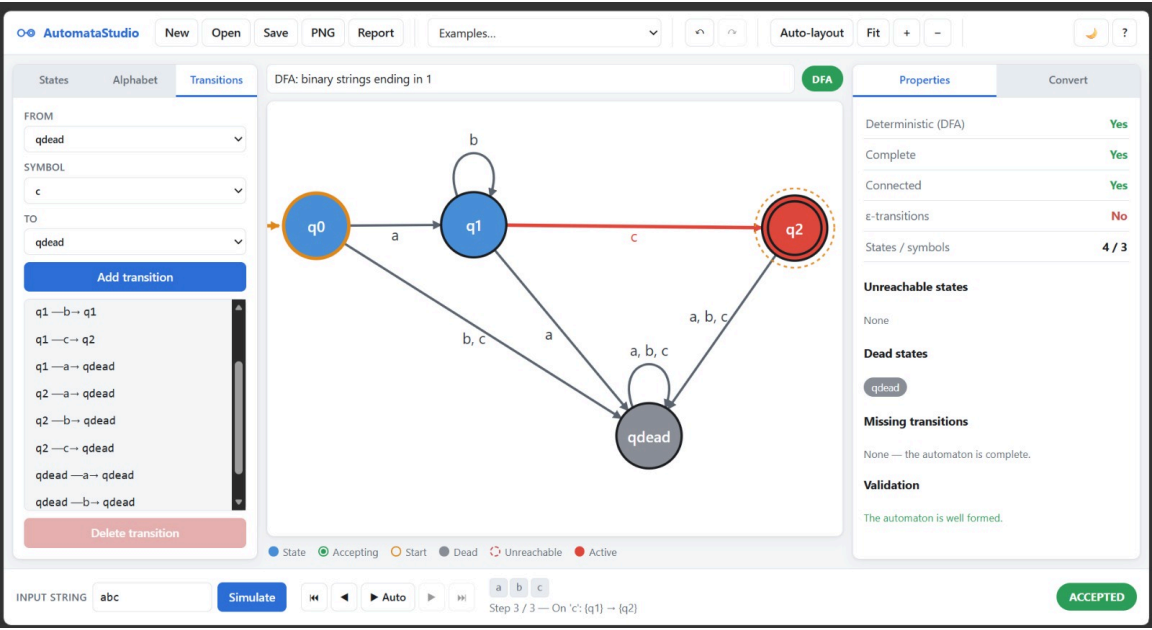
The automaton is well formed.

INPUT STRING ac | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

a c

Step 2 / 2 — On 'c': {q1} → {q2}

ACCEPTED


AutomataStudio
New
Open
Save
PNG
Report
Examples...
Auto-layout
Fit
+
−
?
States
Alphabet
Transitions
DFA: binary strings ending in 1
DFA
Properties
Convert
FROM
qdead
SYMBOL
c
TO
qdead
Add transition
q1 —b→ q1
q1 —c→ q2
q1 —a→ qdead
q2 —a→ qdead
q2 —b→ qdead
q2 —c→ qdead
qdead —a→ qdead
qdead —b→ qdead
Delete transition
b
q0
a
q1
c
q2
b, c
a
a, b, c
a, b, c
qdead
Deterministic (DFA)
Yes
Complete
Yes
Connected
Yes
ε-transitions
No
States / symbols
4 / 3
Unreachable states
None
Dead states
qdead
Missing transitions
None — the automaton is complete.
Validation
The automaton is well formed.
State
Accepting
Start
Dead
Unreachable
Active
INPUT STRING
abc
Simulate
▶ Auto
a
b
c
Step 3 / 3 — On 'c': {q1} → {q2}
ACCEPTED

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

FROM
q3

SYMBOL
c

TO
q4

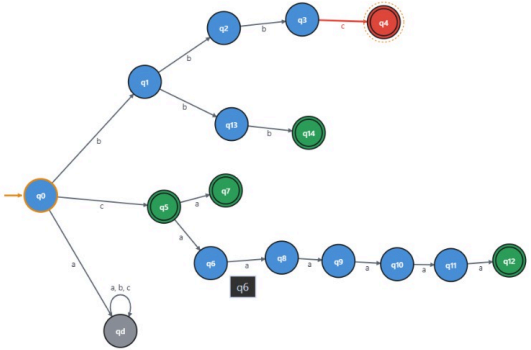
Add transition

q1 —b→ q2
q2 —b→ q3
q3 —c→ q4
q5 —a→ q6
q5 —a→ q7
q6 —a→ q8
q8 —a→ q9
q9 —a→ q10

Delete transition

Untitled automaton

NFA



State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

Unreachable states

None

Dead states

qd

Missing transitions

- q1 --a--> (undefined)
- q1 --c--> (undefined)
- q2 --a--> (undefined)
- q2 --c--> (undefined)
- q3 --a--> (undefined)
- q3 --b--> (undefined)
- q4 --a--> (undefined)
- q4 --b--> (undefined)
- q4 --c--> (undefined)
- q5 --b--> (undefined)
- q5 --c--> (undefined)
- q6 --b--> (undefined)

...and 20 more.

Validation

The automaton is well formed.

INPUT STRING bbbc | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

b b b c

Step 4 / 4 — On 'c': {q14, q3} → {q4}

ACCEPTED

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

Untitled automaton | NFA

Properties | Convert

FROM
q3

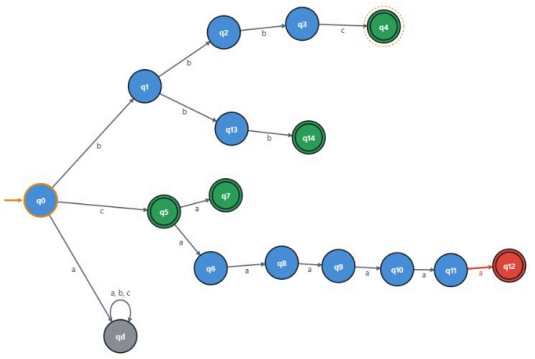
SYMBOL
c

TO
q4

Add transition

q1 —b→ q2
q2 —b→ q3
q3 —c→ q4
q5 —a→ q6
q5 —a→ q7
q6 —a→ q8
q8 —a→ q9
q9 —a→ q10

Delete transition



State | Accepting | Start | Dead | Unreachable | Active

Unreachable states

None

Dead states

qd

Missing transitions

- q1 --a--> (undefined)
- q1 --c--> (undefined)
- q2 --a--> (undefined)
- q2 --c--> (undefined)
- q3 --a--> (undefined)
- q3 --b--> (undefined)
- q4 --a--> (undefined)
- q4 --b--> (undefined)
- q4 --c--> (undefined)
- q5 --b--> (undefined)
- q5 --c--> (undefined)
- q6 --b--> (undefined)

...and 20 more.

Validation

The automaton is well formed.

INPUT STRING caaaaaa | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

c a a a a a a a

Step 7 / 7 — On 'a': {q11} → {q12}

ACCEPTED